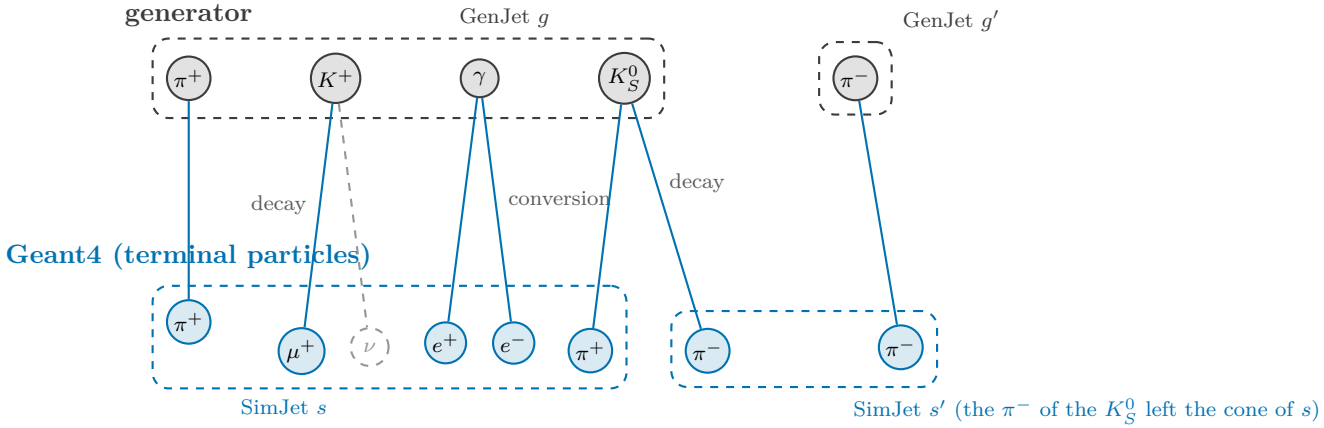


every SimTrack has one generator ancestor (walk `SimTrack_parentIdx` up to the primary) **every generator particle sits in at most one GenJet** (gen jets re-clustered in the job)

for a pair (s, g): $E_{\text{shared}} = \sum_{c \in s,\ \text{anc}(c) \in g} E_c = E_{\pi^+} + E_{\mu^+} + E_{e^+} + E_{e^-} + E_{\pi^+}$

`simFraction` $= E_{\text{shared}}/E(s)$: purity of s `genFraction` $= \dfrac{1}{E(g)} \sum_{p \in g} E_p\, f_s(p)$: efficiency of

g into s, with $f_s(K^0_S) = \dfrac{E_{\pi^+}}{E_{\pi^+} + E_{\pi^-}}$, $f_s(K^+) = 1$ (the ν is not clustered)

matched: the best pair for g with both fractions > 0.5; `ptRatio` $= p_T(s)/p_T(g)$ is the response of the pair

no ΔR anywhere: immune to the bending in the field and to the decay kinematics at low p_T